

Homework 2_Align Reads by Bowtie2

I downloaded **RNA-seq paired-end reads** (Reads1: **ENCFF002DKA**, Reads2: **ENCFF002DKE**) for the **K562 cell line** from the **ENCODE project** (Experiment **ENCSR109IQO**, Library **ENCLB822JYE**). These **FASTQ files** contain raw sequencing data for further analysis. The download was completed as part of **Homework1**,

Quality Control: Assessing the quality of the RNA-seq data to ensure it's suitable for alignment.

1. **Alignment:** Aligning the reads to a reference genome (e.g., **hg19** or other appropriate genomes).
2. **Expression Analysis:** Quantifying gene expression based on the aligned reads, such as identifying differentially expressed genes or exploring gene-level expression patterns.

```
Mohana_Alla 🤔🤔🤔🤔 ~ % brew upgrade Bowtie2
==> Auto-updating Homebrew...
Adjust how often this is run with HOMEBREW_AUTO_UPDATE_SECS or disable with
HOMEBREW_NO_AUTO_UPDATE. Hide these hints with HOMEBREW_NO_ENV_HINTS (see `man brew`)
.
==> Auto-updated Homebrew!
Updated 2 taps (homebrew/core and homebrew/cask).
==> New Formulae
ad                               gdtoolkit                       pgbackrest
aqtinstall                      ggh                             pgrx
behaviortree.cpp                gnome-builder                   postgresql-hll
catgirl                         gomi                            ramalama
cfnc1                            hcledit                         rink
cloud-provider-kind             hypopg                          scryer-prolog
comrak                          icann-rdap                      sdl3_image
dud                             lazysql                         text-embeddings-inference
evil-helix                      libcdio-paranoia                yices2
gauth                           martin
==> New Casks
chatwise                        ik-product-manager              thumbhost3mf
cherry-studio                  open-eid                        turbotax-2024
gpt4all                         oracle-jdk-javadoc@21           vernier-spectral-analysis

You have 15 outdated formulae installed.

==> Upgrading 1 outdated package:
bowtie2 2.5.3 -> 2.5.4
==> Downloading https://ghcr.io/v2/homebrew/core/bowtie2/manifests/2.5.4
##### 100.0%
==> Fetching bowtie2
==> Downloading https://ghcr.io/v2/homebrew/core/bowtie2/blobs/sha256:3f8eca3ff38d6b5
##### 100.0%
==> Upgrading bowtie2
```

```

Mohana_Alla 🐼🐼🐼Bio_App_new % cd ..
Mohana_Alla 🐼🐼🐼Desktop % cd ..
Mohana_Alla 🐼🐼🐼~ % bowtie2 -x ./hg19/hg19 -1 ENCF002DKA.fastq.gz -2 ENCF002DKE.fastq.gz -S ENCF002D
KA_DKE_Aligned.sam
stat: Bad file descriptor
Warning: Could not open read file "ENCF002DKA.fastq.gz" for reading; skipping...
Error: No input read files were valid
(ERR): bowtie2-align exited with value 1
Mohana_Alla 🐼🐼🐼~ % bowtie2 -x ./hg19/hg19 -1 ENCF002DKA.fastq -2 ENCF002DKE.fastq -S ENCF002DKA_DKE
_Aligned.sam
74096905 reads; of these:
  74096905 (100.00%) were paired; of these:
    18447072 (24.90%) aligned concordantly 0 times
    31206255 (42.12%) aligned concordantly exactly 1 time
    24443578 (32.99%) aligned concordantly >1 times
    ----
    18447072 pairs aligned concordantly 0 times; of these:
      4038037 (21.89%) aligned discordantly 1 time
    ----
    14409035 pairs aligned 0 times concordantly or discordantly; of these:
      28818070 mates make up the pairs; of these:
        18055514 (62.65%) aligned 0 times
        7068549 (24.53%) aligned exactly 1 time
        3694007 (12.82%) aligned >1 times
87.82% overall alignment rate
Mohana_Alla 🐼🐼🐼~ % █

```

I installed **Bowtie2** and downloaded the **hg19 reference genome**. After then, I aligned **paired-end RNA-seq reads (ENCF002DKA & ENCF002DKE)** to the **hg19 genome** using the following command:

```
bowtie2 -x ./hg19/hg19 -1 ENCF002DKA.fastq.gz -2 ENCF002DKE.fastq.gz
-S ENCF002DKA_DKE_Aligned.sam
```

This command performed the following:

- **-x ./hg19/hg19**: Specifies the **hg19 reference genome** index for alignment.
- **-1 ENCF002DKA.fastq.gz**: Input for **Read1**.
- **-2 ENCF002DKE.fastq.gz**: Input for **Read2**.
- **-S ENCF002DKA_DKE_Aligned.sam**: Output in **SAM format**, which contains the alignment results.

This process aligned your RNA-seq reads to the **hg19 reference genome** and generated a **SAM file** for further analysis.

```

zsh: command not found: -zsh
Mohana_Alla 🍌🍌🍌~ % samtools --version
bcftools --version
htslib --version

samtools 1.21
Using htslib 1.21
Copyright (C) 2024 Genome Research Ltd.

Samtools compilation details:
Features:      build=configure curses=yes
CC:           clang
CPPFLAGS:
CFLAGS:       -Wall -g -O2
LDFLAGS:
HTSDIR:
LIBS:
CURSES_LIB:   -lcurses

HTSlib compilation details:
Features:      build=configure libcurl=yes S3=yes GCS=yes libdeflate=yes lzma=yes bzip2=yes plugins=no htscdec=1.6.1
CC:           clang
CPPFLAGS:
CFLAGS:       -Wall -g -O2 -fvisibility=hidden
LDFLAGS:      -fvisibility=hidden

HTSlib URL scheme handlers present:
built-in:     file, preload, data
S3 Multipart Upload:  s3w+https, s3w+http, s3w
Amazon S3:    s3+https, s3, s3+http
Google Cloud Storage:  gs+http, gs+https, gs
libcurl:      gophers, smtp, ldaps, smb, rtsp, ftp, pop3, smbs, imaps, pop3s, ftps, https, http, ftp, gopher, imap, ldap, smtps, dict, mqtt, telnet
crypt4gh-needed:  crypt4gh
mem:          mem

bcftools 1.21
Using htslib 1.21
Copyright (C) 2024 Genome Research Ltd.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
zsh: command not found: htslib
Mohana_Alla 🍌🍌🍌~ %

```

successfully **downloaded and installed Samtools, Bcftools, and HTSlib** from the official websites. These tools are crucial for **processing genomic data**, including:

- **Samtools:** Used for handling **SAM/BAM files**, such as conversion between SAM and BAM formats, sorting, indexing, and manipulating alignment data.
- **Bcftools:** A set of tools for **variant calling and manipulation of VCF (Variant Call Format) files**, which are used to store information about genetic variants.
- **HTSlib:** A C library designed for reading and writing **high-throughput sequencing data files** (e.g., BAM, CRAM, VCF).

After installation, you **verified the setup** of these tools using the following commands:

For Samtools:

```
samtools --version
```

For Bcftools:

```
bcftools --version
```

For HTSlib:

```
htslib --version
```

These commands will help to check the successful installation of the tools, making them ready for use in genomic data processing, alignment, variant calling, and other bioinformatics analyses.

```
Mohana_Alla 🤔🤔🤔🤔 % samtools view -bS ENCFF002DKA_DKE_Aligned.sam > ENCFF002DKA_DKE_Aligned.bam
Mohana_Alla 🤔🤔🤔🤔 % samtools sort ENCFF002DKA_DKE_Aligned.bam -o ENCFF002DKA_DKE_Aligned_Sorted.bam
[bam_sort_core] merging from 27 files and 1 in-memory blocks...
Mohana_Alla 🤔🤔🤔🤔 % samtools index ENCFF002DKA_DKE_Aligned_Sorted.bam
```

After aligning the RNA-seq reads to the **hg19** reference genome using **Bowtie2**, the output was a **SAM file**(**ENCFF002DKA_DKE_Aligned.sam**). Since SAM files are large and inefficient for processing, we convert them to **BAM format**, which is a compressed and indexed binary version.

Step 1: Convert SAM to BAM

Run the following command to convert the alignment file from **SAM to BAM** format:

```
samtools view -bS ENCFF002DKA_DKE_Aligned.sam >
ENCFF002DKA_DKE_Aligned.bam
```

Explanation:

- **samtools view** → Extracts data from a SAM/BAM file.
- **-b** → Specifies BAM output format.
- **-S** → Indicates input is a SAM file.
- **ENCFF002DKA_DKE_Aligned.sam** → Input SAM file.
- **ENCFF002DKA_DKE_Aligned.bam** → Output BAM file.

At this stage, the file is still **unsorted**, which is inefficient for downstream processing.

Step 2: Sort the BAM File

Sorting organizes reads by reference position, making the file more efficient for indexing and variant calling.

```
samtools sort ENCFF002DKA_DKE_Aligned.bam -o
ENCFF002DKA_DKE_Aligned_Sorted.bam
```

Explanation:

- **samtools sort** → Sorts BAM files.
- **ENCFF002DKA_DKE_Aligned.bam** → Input BAM file.
- **-o ENCFF002DKA_DKE_Aligned_Sorted.bam** → Specifies the output sorted BAM file.

The output file, **ENCFF002DKA_DKE_Aligned_Sorted.bam**, is now optimized for indexing and visualization.

Step 3: Index the Sorted BAM File

Indexing enables rapid access to specific regions in the alignment file, necessary for visualization tools like **IGV** and variant calling.

```
samtools index ENCFF002DKA_DKE_Aligned_Sorted.bam
```

Explanation:

- **samtools index** → Creates an index for BAM files.
- **ENCFF002DKA_DKE_Aligned_Sorted.bam** → Input sorted BAM file.
- The output is **ENCFF002DKA_DKE_Aligned_Sorted.bam.bai**, an index file used for fast access.

Final Output Files:

1. **ENCFF002DKA_DKE_Aligned.bam** → Unsorted BAM file.
2. **ENCFF002DKA_DKE_Aligned_Sorted.bam** → Sorted BAM file.
3. **ENCFF002DKA_DKE_Aligned_Sorted.bam.bai** → Index file for efficient retrieval.